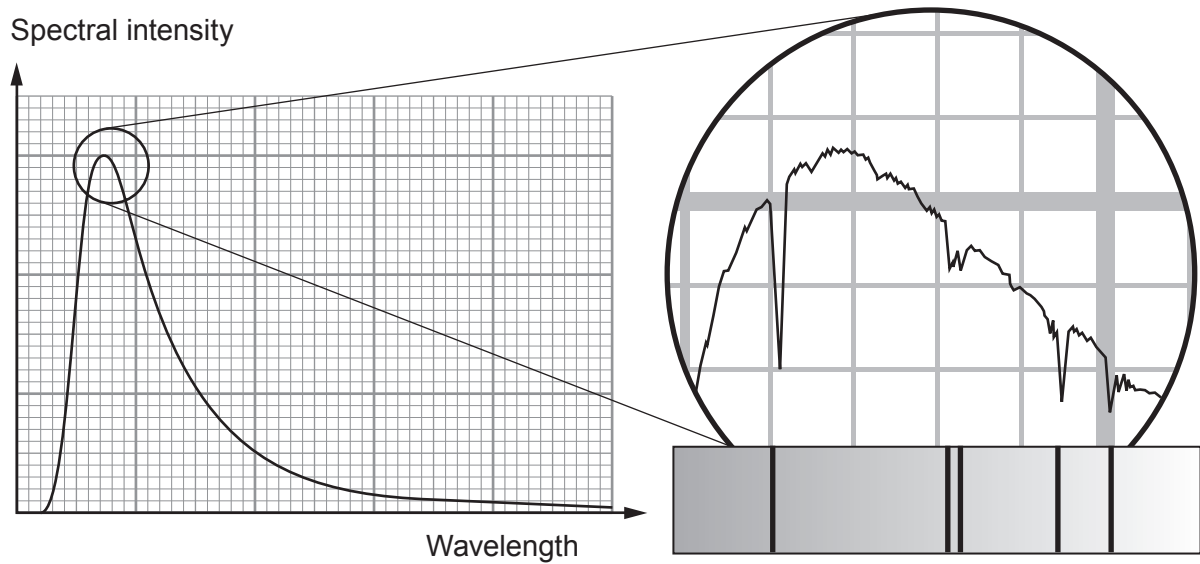


Examiner
only

6. (a) A blackbody graph of spectral intensity against wavelength for a star is shown. A magnified section, showing the finer detail of the spectrum is also given. An associated line spectrum is also shown.



Explain how the graph and the spectra can be used to provide information about the star and the elements from which it is made. [6 QER]



Examiner
only

- (b) (i) Altair is the brightest star in the Aquila constellation. It is 1.58×10^{17} m away, and the intensity of its electromagnetic radiation reaching the Earth is $1.32 \times 10^{-8} \text{ W m}^{-2}$. Show that its luminosity is approximately $4 \times 10^{27} \text{ W}$. [3]

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- (ii) Calculate Altair's **diameter** given that its surface temperature is 7 700 K. [3]

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12

TURN OVER FOR THE
LAST QUESTION

